Practice workbook · Spot selection and iteration, and pick the right loop

Naming the construct shown in a flowchart or keyword, identifying the kind of loop in an algorithm, and explaining why a condition-controlled loop suits a task better than a count-controlled one.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: CSC-T2-06

1. Question 6 · part b.iv · 2 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part b.iv only.

6 Algorithms can be represented using flowcharts or pseudocode.

(a) An algorithm has been designed to perform these steps:

- A user is asked to input a number **n**
- If the number **n** is 0 the program stops
- If the number **n** is positive a message to display 'Positive' is output
- If the number **n** is negative a message to display 'Negative' is output
- The program repeats until a value of 0 has been input.

Draw a flowchart to represent this algorithm.

(6)

OLevels.COM — SCHOOL —

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

16

P 7 8 6 3 7 A 0 1 6 2 0

1. Question 6 · part b.iv · 2 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part b.iv only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

(b) **Figure 3** shows an algorithm written in pseudocode that performs a mathematical process.

```
01
02  RECEIVE number FROM (INTEGER) KEYBOARD
03
04  SET flag TO FALSE
05
06  IF number = 1 THEN
07     SET flag TO TRUE
08  ELSE
09     SET testNum TO 1
10     REPEAT
11        SET testNum TO testNum + 1
12        IF (number MOD testNum) = 0 THEN
13           SET flag TO TRUE
14        END IF
15     UNTIL (testNum = number) OR (flag = TRUE)
16  END IF
17
18  IF NOT (flag = TRUE)
19     SEND number TO DISPLAY
20  END IF
21
```

Figure 3

Trace tables are used to show the outputs from algorithms.

(i) Complete the trace table for an input of 7

(2)

number	flag	testNum	DISPLAY

OLevels.com SCHOOL

1. Question 6 · part b.iv · 2 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part b.iv only.

(ii) Complete the trace table for an input of 9

(2)

number	flag	testNum	DISPLAY

(iii) State the purpose of the algorithm.

(1)

(iv) Lines 10 to 15 of the pseudocode in **Figure 3** use a post-conditioned loop.

Explain **one** reason why it is better for the algorithm to use a post-conditioned loop rather than a count controlled loop.

(2)

(Total for Question 6 = 13 marks)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN

OLevels.com SCHOOL

1. Question 6 · part b.iv · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 1 · part c · 1 mark

2021 NOV · Paper 02

Further practice · Attempt part c only.

1 Programmers write code to solve problems.

(a) Give **one** technique that programmers use to make their code easy to read and understand.

(1)

.....

(b) Programs may contain logic, syntax or runtime errors.

A program has three errors.

These may be logic, syntax or runtime errors.

Complete the table by adding a tick (✓) for **each** error description to show whether the error is a logic, syntax or runtime error.

(3)

Error description	Logic	Syntax	Runtime
Divide by 0			
Use x instead of * to multiply			
Subtract 10 from 2 instead of 2 from 10			

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

2. Question 1 · part c · 1 mark

2021 NOV · Paper 02

Further practice · Attempt part c only.

(c) **Figure 1** shows a programming construct in a flowchart.

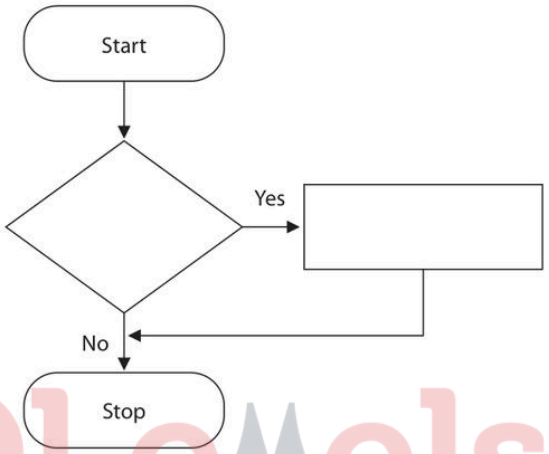


Figure 1

Identify the name of the programming construct.

(1)

- ☒ **A** Array
- ☒ **B** Iteration
- ☒ **C** Operator
- ☒ **D** Selection

(d) Open **Q01d** in the code editor.

The program inputs two numbers, compares them and outputs a suitable message.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01dFINISHED** with the correct file extension for the programming language.

(3)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

2. Question 1 · part c · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 6 · part a · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part a only.

6 Pseudocode and flowcharts are used to create algorithms.

(a) The number of minutes a train is early or late are recorded each day for a week.

- A value of 0 is recorded if the train is on time.
- A negative value is recorded if the train is early.
- A positive value is recorded if the train is late.

The algorithm in **Figure 2** has been created to calculate and output the total number of trains that were early, on time or late.

```
01
02  SET arrival TO [-2, 1, 5, 0, -3, 4, 1]
03
04  SET early TO 0
05  SET late TO 0
06  SET index TO 0
07
08  WHILE NOT (index > LENGTH(arrival)) DO
09
10      IF arrival[index] >= 0 THEN
11          SET late TO arrival[index]
12      ELSE
13          IF arrival[index] < 0 THEN
14              SET early TO early + 1
15          END IF
16      END IF
17
18      SET index TO index + 1
19
20  END WHILE
21
22  SEND "Trains early:    " & early TO DISPLAY
23  SEND "Trains on time: " & (late + early) TO DISPLAY
24  SEND "Trains late:     " & late TO DISPLAY
25
```

Figure 2

Identify the type of iteration construct used in **Figure 2**.

(1)

- ☒ **A** Post-conditioned loop
- ☒ **B** Count controlled loop
- ☒ **C** Array loop
- ☒ **D** Pre-conditioned loop

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

22

P 7 5 7 3 6 A 0 2 2 2 8

3. Question 6 · part a · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part a only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(b) There are **four** errors in the pseudocode in **Figure 2**.

Write the correct code for lines 08, 10, 11 and 23

(i) Line 08

(1)

(ii) Line 10

(1)

(iii) Line 11

(1)

(iv) Line 23

(1)

(c) Give **one** reason for the variable `index` being incremented in line 18.

(1)

OLevels.com SCHOOL

3. Question 6 · part a · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 1 · part a · 1 mark

2025 JUN · Paper 02

Further practice · Attempt part a only.

1 Programmers use different programming constructs to create working code.

(a) Identify the keyword used to create a selection statement.

(1)

- ☒ **A** for
- ☒ **B** if
- ☒ **C** new
- ☒ **D** while

(b) Computer code can contain errors.

(i) Complete the table by adding a tick (✓) to match each error description to a type of error.

(3)

Description	Syntax error	Logic error	Runtime error
The program does not produce the expected output.			
The program does not translate.			
The program crashes during execution.			

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

4. Question 1 · part a · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL